

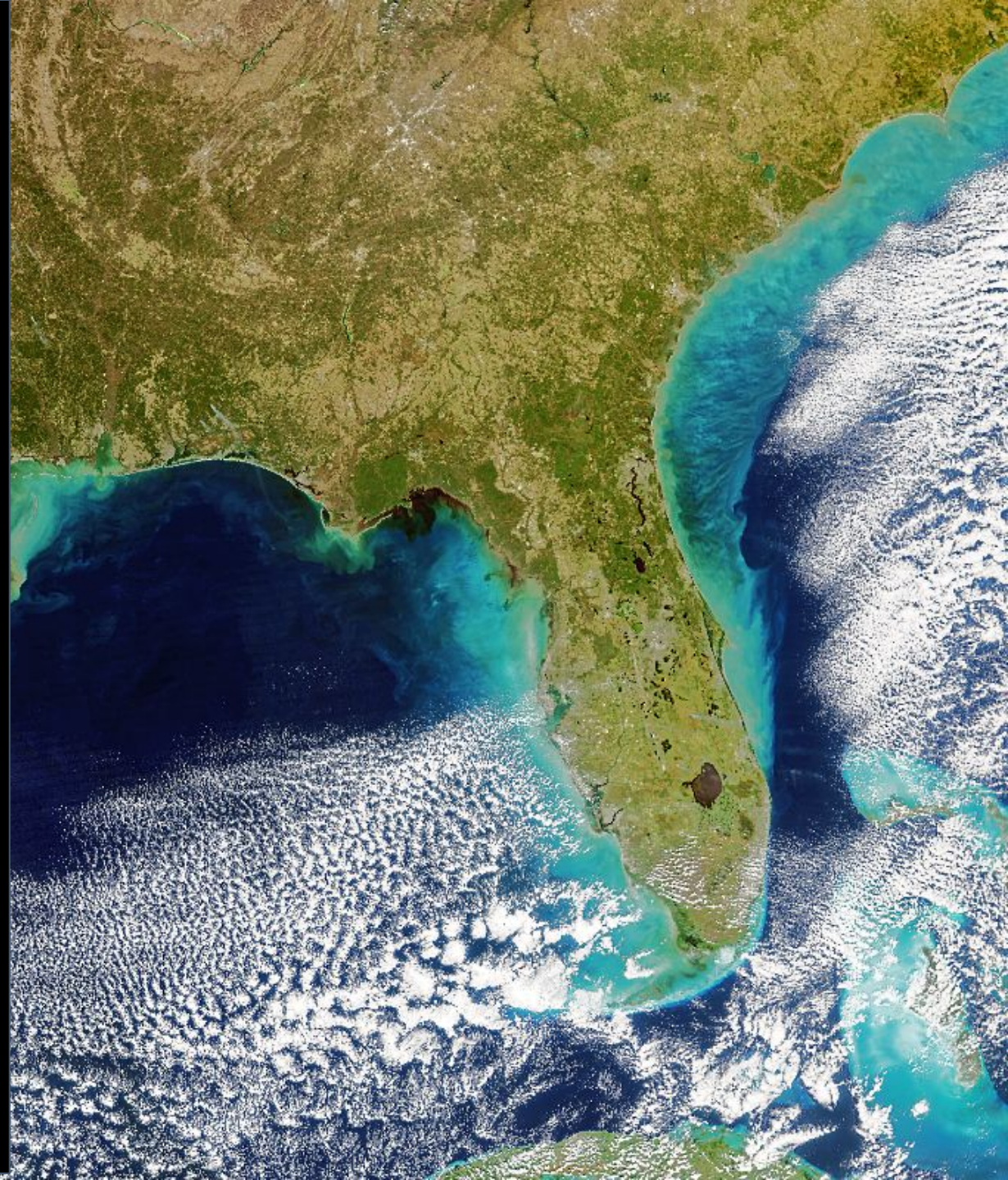
Introduction to PACE hyperspectral ocean color data for fisheries applications

Eli Holmes, Climate/OA Monthly mtg
8 April 2025

Thanks to

Jeremy Werdell, PACE Project Scientist, NASA
Goddard for PACE slides

Ryan Vandermuelen, NOAA Fisheries Satellite
Coordinator for fisheries applications slides



Late 2014 PACE became a mission and work began early 2015



LAUNCH 8 February 2024
06:33UTC

Picture, NASA

NASA Plankton, Aerosol, Cloud, ocean Ecosystem (PACE) mission

PACE will extend **key systematic ocean color, aerosol, & cloud (& land) climate data records**.

PACE will reveal the **diversity of organisms fueling marine food webs** & how ecosystems respond to change.

Looking at the ocean, clouds, land & aerosols together will improve knowledge of the roles each plays in our planet.

The observatory:

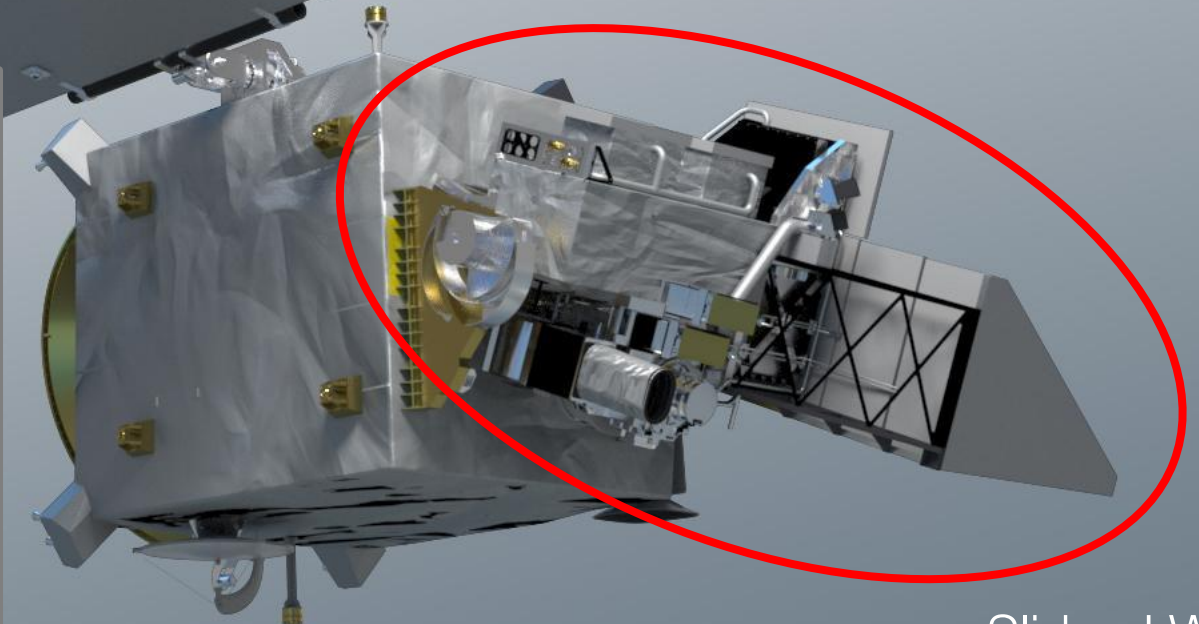
- Ocean Color Instrument (primary) – NASA GSFC
- SPeXone – SRON/Airbus (Netherlands)
- HARP2 – University of Maryland Baltimore County
- Mission Control – NASA GSFC



ocean color & the ocean color instrument

ocean color retrievals drive OCI's
design & performance requirements

- hyperspectral scanning radiometer
- (320) 340 – 890 nm, 5 nm resolution, 2.5 nm steps⁺
- plus, 940, 1038, 1250, 1378, 1615, 2130, and 2250 nm
- *single science pixel to mitigate image striping*
- 1 – 2 day global coverage
- ground pixel size of 1 km² at nadir
- ± 20° fore/aft tilt to avoid Sun glint
- twice monthly lunar calibration
- daily on-board solar calibration
- <0.5% total system error for VIS-NIR
- SNRs optimized for ocean color science
- [simulated top-of-atmosphere data available](#)



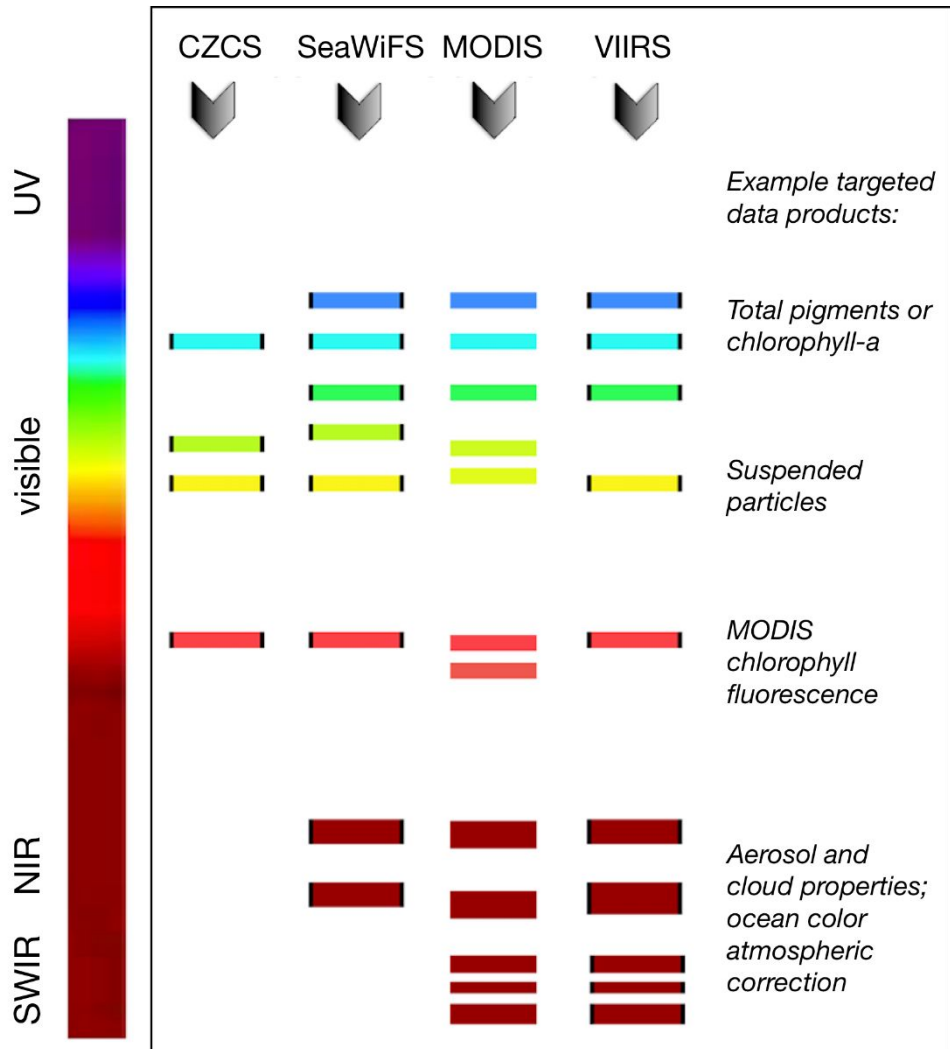
+ with 1.25 nm steps in several spectral regions

* developed primarily for mechanical processing assessments

Slide, J Werdell

moving from multi-spectral radiometry to spectroscopy

1978-1986 1997-2010 1999-pres. 2012-pres.



Source: <https://pace.gsfc.nasa.gov>.



Example diatom



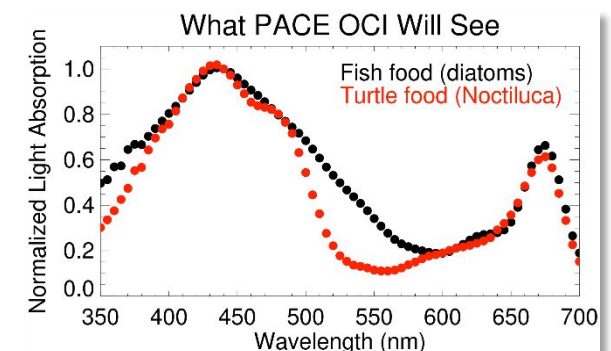
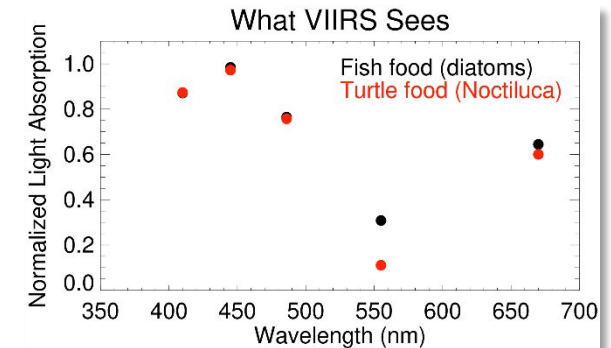
Linda Armbricht, abc.com.au

Example Noctiluca



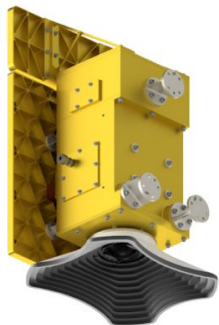
1 mm
Joaquim Goes, LDEO

signals from the ocean are small & differentiating between constituents requires additional information relative to what we have today

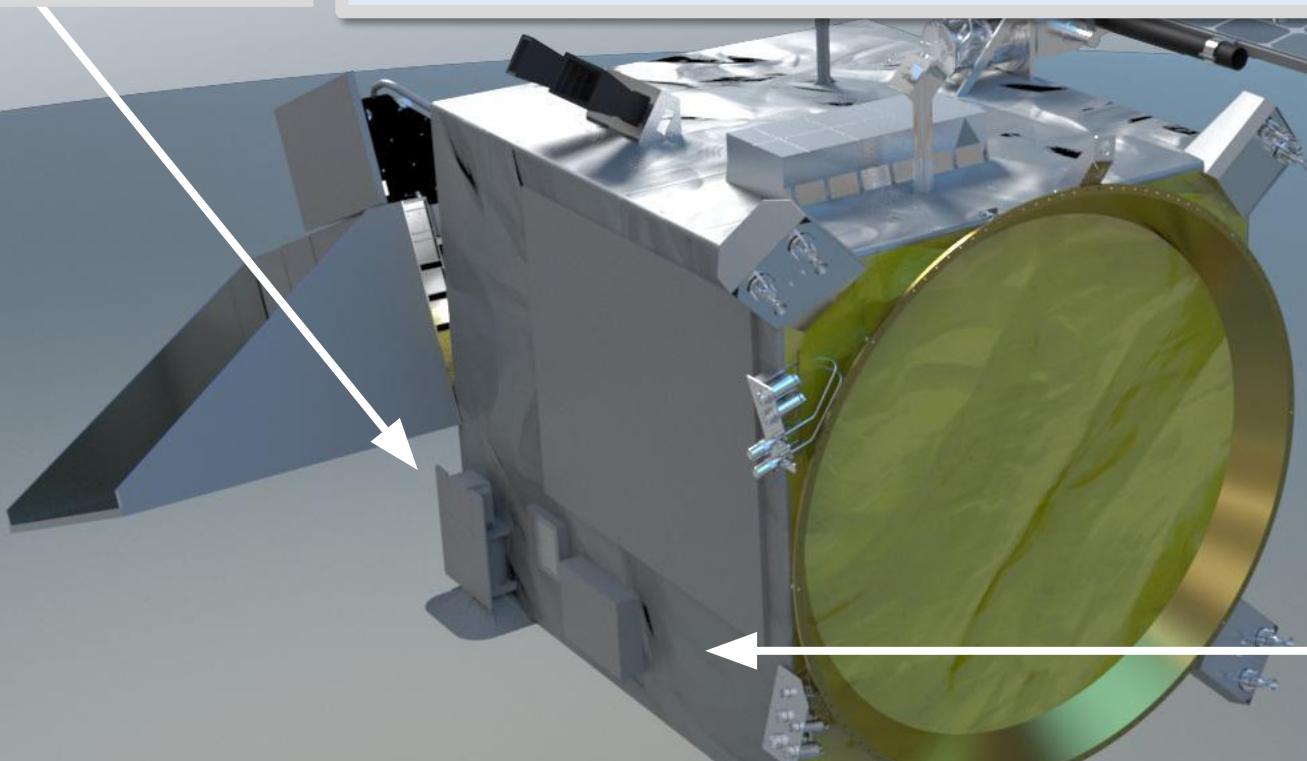


Slide, J Werdell

UMBC Hyper
Angular
Rainbow
Polarimeter
(HARP-2)



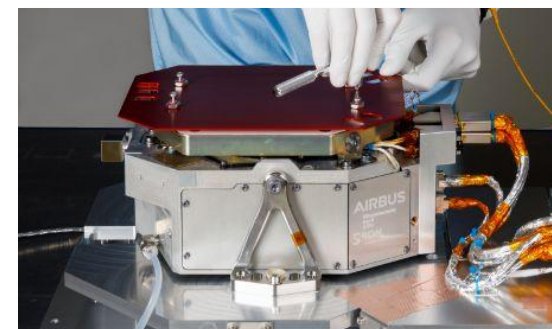
- Excellent for cloud droplet size and ice particle shape/roughness retrievals
- Wide swath matches OCI, offering potentially improved atmos. Correction*
- Physical properties of aerosols: size, shape, distribution vertically and horizontally, how much light do they block (opaqueness)*



Slide modified from, J Werdell

- Aerosol characterization
- What are they made of? Sulfates, dust, smoke, pollution etc*
- How do they affect light? Refraction, absorption, scattering*

SRON/Airbus
Spectro-polarimeter for
Planetary
Exploration
(SPEXone)



Calibrated Radiometry and Polarimetry | Ocean Properties to be Produced by OCI | Atmospheric Properties to be Produced by OCI | Land Data Products to be Produced by OCI | Aerosol and Ocean Properties from HARP2 | Aerosol and Land Surface Properties from HARP2 | Cloud Properties from HARP2 | Ocean Surface Properties from HARP2 | Aerosol and Ocean Properties from SPEXone | Aerosol and Land Surface Properties from SPEXone | Aerosol and Ocean Properties from OCI + HARP2 + SPEXone

Access to data varies with its status (data maturity level). Provisional data are available through [Earthdata Search](#), the [OB.DAAC File Search](#) and [Level 3 & 4 Browser](#). Test and Diagnostic data are available through the [OB.DAAC File Search](#) and [Level 3 & 4 Browser](#). See also “[Access PACE Data](#)”.

What do colors in the “Availability” column mean?

Available

Coming soon!

Currently implementing and evaluating

No approach currently identified

Calibrated Radiometry and Polarimetry					
Calibrated and geolocated radiometry and polarimetry as observed at sensor.					
Product	Description and Use	Units	Availability	Status	Additional Info
Spectral top-of-atmosphere radiances from OCI	Spectral radiance observed at the top of the atmosphere.	$\text{W m}^{-2} \text{um}^{-1} \text{sr}^{-1}$	Level-1B 1-km at nadir; daily - Level-1C ; daily	Provisional	Level-1C draft data format and examples
Spectral top-of-atmosphere radiances and polarimetry from SPEXone	Spectral radiance and polarimetry observed at the top of the atmosphere, for all sensor viewing angles.	Various	Level-1B TBD; daily - Level-1C ; daily	Provisional	Level-1C draft data format and examples
Spectral top-of-atmosphere radiances and polarimetry from HARP2	Spectral radiance and polarimetry observed at the top of the atmosphere, for all sensor viewing angles.	Various	Level-1B TBD; daily - Level-1C ; daily	Provisional	Level-1C draft data format and examples

Ocean Properties to be Produced by OCI					
Bio-optical and biogeochemical properties of seawater constituents in the sunlit upper ocean.					
Product	Description and Use	Units	Availability	Status	Additional Info
Spectral remote sensing reflectances	Spectral color of the ocean in the ultraviolet-to-near infrared spectral range. Used as input into algorithms to retrieve information about colored dissolved organic matter, phytoplankton, non-algal particles, and other aquatic constituents. Provided in continuous 2.5-nm steps from 350 to 717.5-nm with a resolution (bandwidth) of 5-nm.	sr^{-1}	Level-2 1-km at nadir; daily - Level-3 4-km; daily, 8-day, monthly, annual	Provisional	ATBD SAT members: Boss, Zhai, Krotkov, Chowdhary, Stamnes, Zhang In situ measurement protocols
Apparent visible wavelength	An optical water classification index reported as the weighted harmonic mean of visible-range Rrs wavelengths (400-700 nm)	nm	Level-2 1-km at nadir; daily - Level-3 4-km; daily, 8-day, monthly, annual	Test	ATBD
Spectral diffuse attenuation coefficients	Spectral diffuse attenuation of downwelling irradiance at multiple wavelengths between 350 and 700 nm. Provides indices of water clarity and light penetration.	m^{-1}	Level-2 1-km at nadir; daily - Level-3 4-km; daily, 8-day, monthly, annual	Test	ATBD SAT members: Boss, Stramski, Odenmaier In situ measurement protocols

Levels

Level 1: Calibrated raw data from sensors. Top of atmosphere radiances from OCI, HARP2, and SPEXone. OCI measures Rrs (Remote Sensing Reflectance): color intensity of the ocean when viewed from above.

Level 2: Derived geophysical variables at the original sensor resolution. Products like remote sensing reflectance (Rrs), absorption properties (derived from Rrs), aerosol properties, cloud properties, chlorophyll estimates (derived from absorption), phytoplankton composition

Level 3: Gridded global maps of those variables, averaged over time (daily, monthly, etc.)

Products

Absorption coefficients: how light is absorbed *within* the water by different substances (like phytoplankton or dissolved organic matter). Derived from R_{rs}

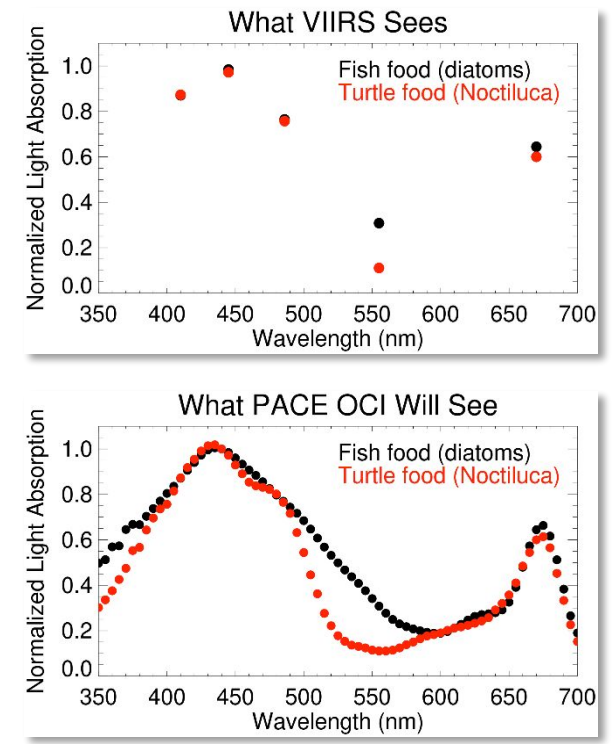
Chlorophyll-a: estimate from empirical equations involving R_{rs} or absorption coefficients

Phytoplankton species and community composition: new to PACE via the hyperspectral absorption coefficients and backscattering coefficients

Diffuse Absorption coefficients (K_d): how quickly light decreases with depth in water at a given wavelength

Spectral phytoplankton absorption coefficients: how much light is absorbed by phytoplankton pigments in water

Phytoplankton carbon: a more direct indicator of phytoplankton biomass



PACE Products for Fisheries, Ryan Vandermuelen (NOAA Fisheries)

PACE products for fisheries and aquaculture



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[Getting started](#) >

[Hyperspectral products](#) >

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Applications of PACE hyperspectral ocean color data for aquaculture and fisheries management

AUTHOR

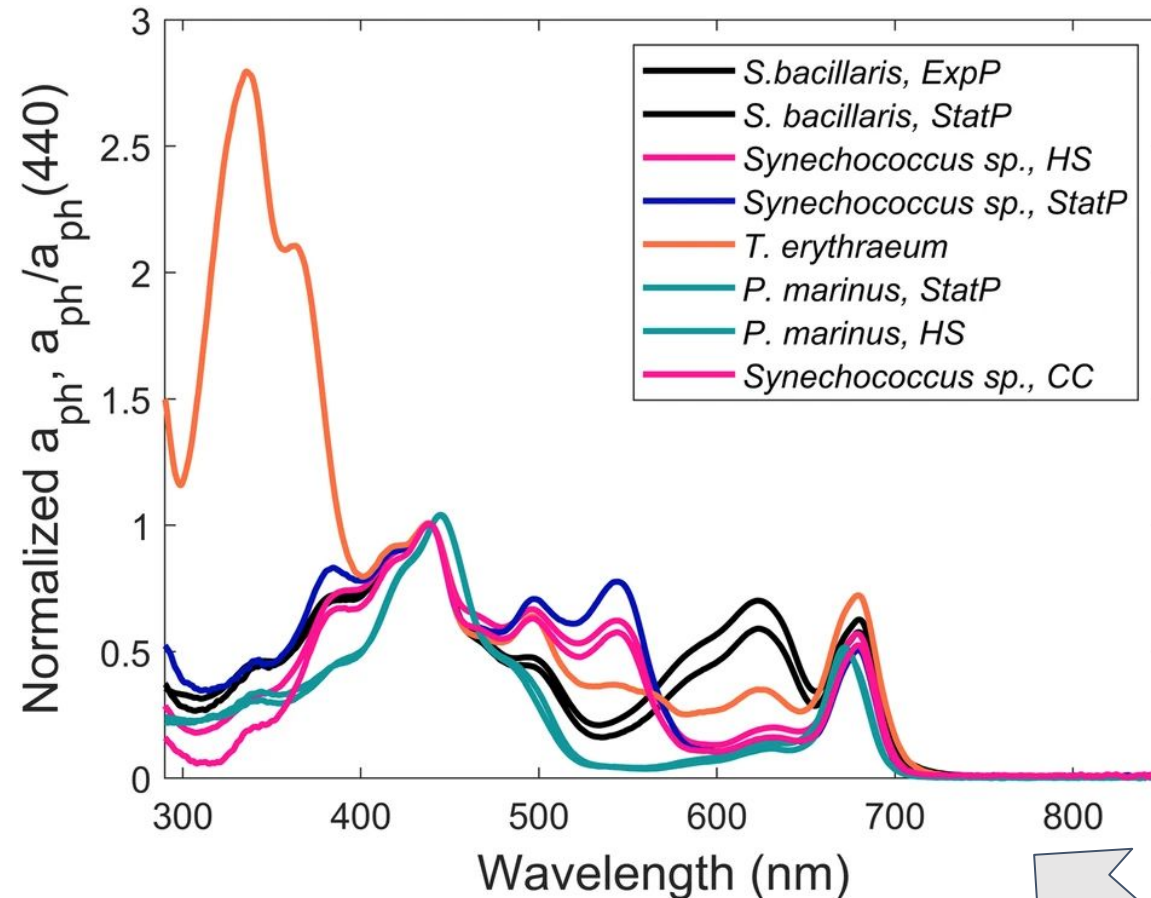
Ryan Vandermulen (NOAA Fisheries, OST), Eli Holmes
(NOAA Fisheries), Others

PUBLISHED

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PACE Ocean products tell you about:

- WHAT IS THE COLOR OF THE OCEAN (Rrs)
- WHAT/WHO IS THERE (phytoplankton composition, SPM)
- HOW MUCH OF IT IS THERE (chlorophyll concentration, particulate/phytoplankton carbon etc.)
- ARE THEY SHINY/COLORFUL??? (IOPs)
- HOW MUCH LIGHT IS THERE (PAR/Kd)
- WHAT ARE THEY DOING (Primary productivity)
- HOW ARE THEY DOING (Fluorescence Line Height)



Review

Phytoplankton composition from sPACE: Requirements, opportunities, and challenges

Ivona Cetinić ^{a b 1}, Cecile S. Rousseaux ^{b 1} ✉, Ian T. Carroll ^{c b}, Alison P. Chase ^d, Sasha J. Kramer ^e, P. Jeremy Werdell ^b, David A. Siegel ^f, Heidi M. Dierssen ^g, Dylan Catlett ^h, Aimee Neeley ^{b i}, Inia M. Soto Ramos ^{a b}, Jennifer L. Wolny ^j, Natasha Sadoff ^{b i}, Erin Urquhart ^{b i}, Toby K. Westberry ^k, Dariusz Stramski ^l, Nima Pahlevan ^{m i}, Bridget N. Seegers ^{a b}, Emerson Sirk ^{b i}, Priscila Kienteca Lange ^{n o} ...Michael Sayers ^t

[nature](#) > [scientific data](#) > [data descriptors](#) > [article](#)

Data Descriptor | [Open access](#) | Published: 03 February 2024

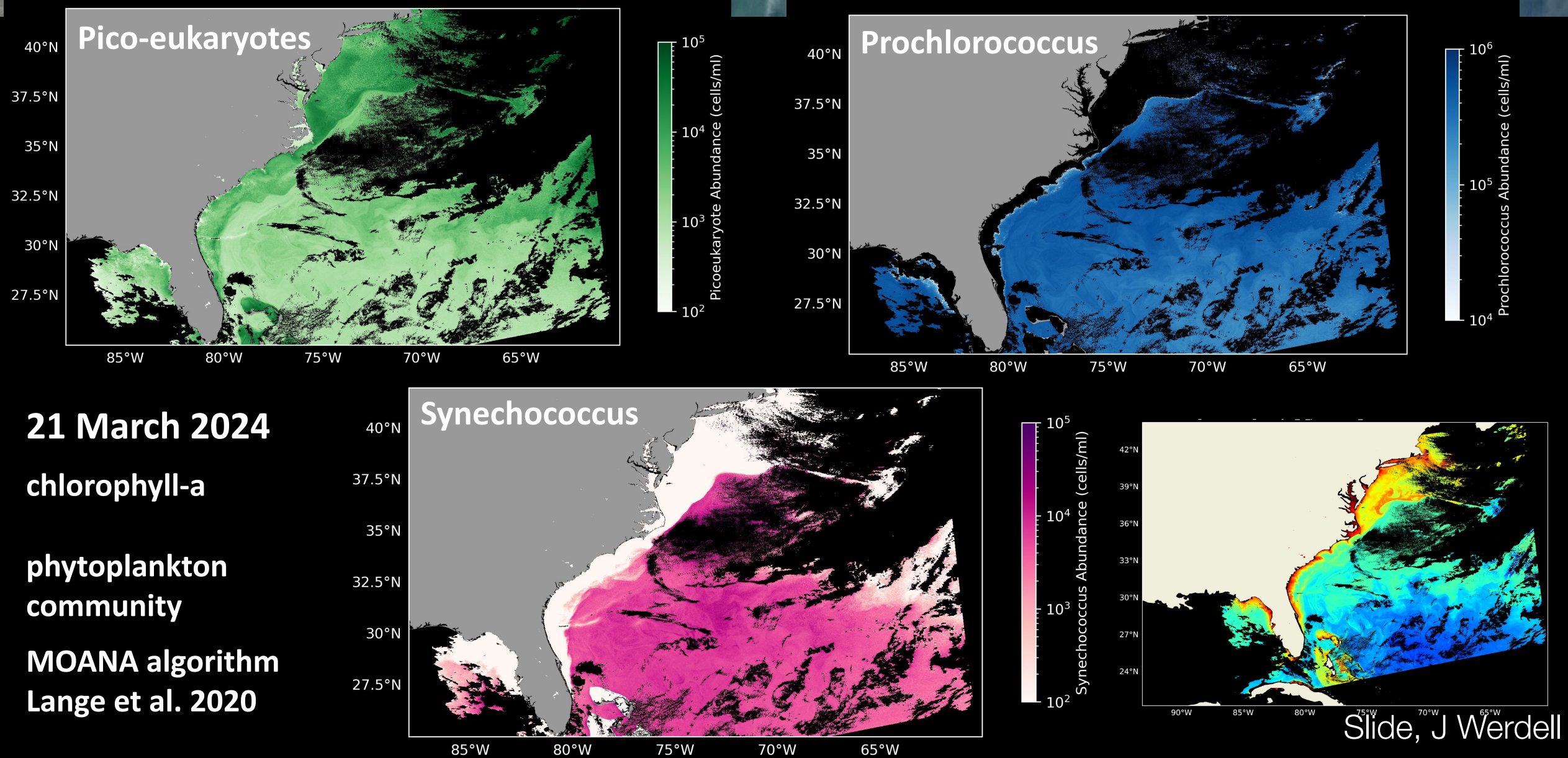
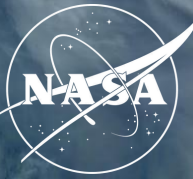
Phytoplankton optical fingerprint libraries for development of phytoplankton ocean color satellite products

Michael W. Lomas ✉, Aimee R. Neeley, Ryan Vandermeulen, Antonio Mannino, Crystal Thomas, Michael G. Novak & Scott A. Freeman

[Scientific Data](#) **11**, Article number: 168 (2024) | [Cite this article](#)

3118 Accesses | 4 Altmetric | [Metrics](#)

PACE OCI Phytoplankton Community Composition



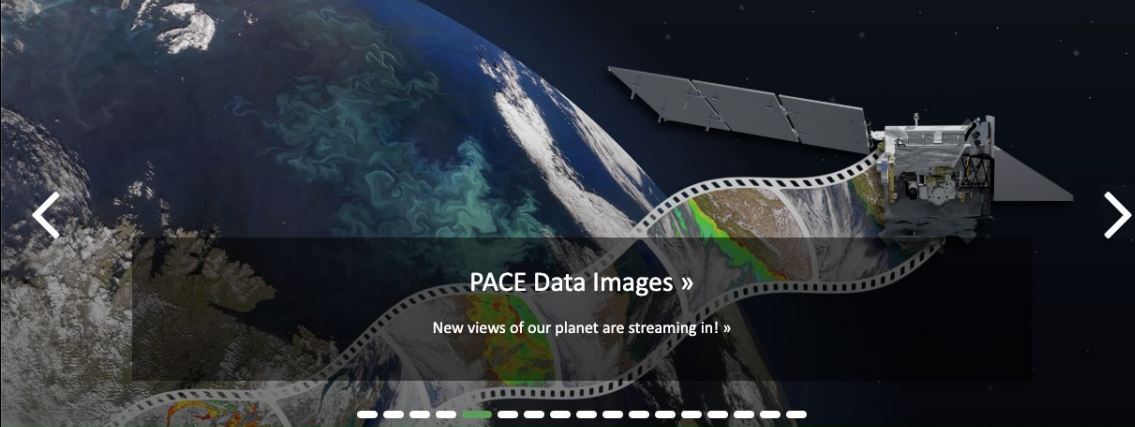
<https://pace.gsfc.nasa.gov>

 National Aeronautics and Space Administration

PACE Plankton, Aerosol, Cloud, ocean Ecosystem



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PACE Data Images »

New views of our planet are streaming in! »

MISSION ELAPSED TIME	DAYS 423	HR 16	MIN 47	ORBIT NUMBER 944	APPROXIMATE POSITION	LAT 24	LON 109	SATELLITE LIGHTING	DAY	CLICK TO SEE ON MAP
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✿ PACE HAS LAUNCHED!

Post-launch UpdatesSatellite Status UpdatesJoin the PACE Community Mailing List

PACE's data is helping us better understand how the ocean and atmosphere exchange carbon dioxide. In addition, it is revealing how aerosols might fuel phytoplankton growth in the surface ocean. Novel uses of PACE data will benefit our economy and society. For example, it will help identify the extent and duration of harmful algal blooms. PACE is extending and expanding NASA's long-term

WHAT'S NEW

02-Apr-25New PACE Terrestrial Data Users' Group→

08-Feb-25PACE-iversary #1→

08-Feb-25A Year of PACE Discoveries→

06-Feb-25PACE-PAX Overview Page→

<https://oceancolor.gsfc.nasa.gov/resources/docs/tutorials/>

 OCEAN COLOR
OB.DAAC | OBPG



ABOUT DATA RESOURCES TOOLS COMMUNITY GALLERY FORUMQuick Links



RESOURCES

HELP HUB

Satellite data processing can be difficult.
We're here to help you climb out of that hole!

We are constantly working on new material. Come back often!

Have an idea? Share it with us on the Earthdata Forum using the tags OB.DAAC under DAAC and Data Recipes under Services/Usage.

Explore our learning resource themes:

- Help Hub Core
- ARSET Training - Introduction to Plankton, Aerosol, Cloud, Ocean Ecosystem (PACE) Hyperspectral Observations for Water Quality Monitoring
- PACE Hackweek 2024
- SeaDAS Basics

Interactive tutorials

Run the code as you follow the training with Jupyter notebooks.

Learn more about Jupyter Notebooks

Cloud computing

Some tutorials use cloud computing resources.

Learn more about the cloud

Help Hub Core

Learn the essential things you need to know to access and process data in this series of Jupyter Notebooks.

Cloud Data Access (OCI Example)

File Structure (OCI Example)

Explore Level 2 Ocean Color Data

Explore Level 3 Ocean Color Data

Installing the SeaDAS Command Line Tools (OCSSW)

Processing with the SeaDAS Command Line Tools (OCSSW)

Come to our EDMW Workshop on working with
PACE data in Python

<https://nosc.noaa.gov/EDMC/edmcon.php>

Check out the PACE HackWeek 2024 website

<https://pacehackweek.github.io/pace-2024/>

Accessing PACE Data

Earthdata Search OB.DAAC portal

Earthdata Search OB.DAAC portal

<https://search.earthdata.nasa.gov/search>

Top left click “Browse Portals”

Click on “OBDAAC”

Filter Instruments to “OCI”

OB.DAAC Search page

<https://oceandata.sci.gsfc.nasa.gov/>

How can I access PACE data that's in the cloud?

You can directly download netcdfs and run analysis. But these files are really big...often better to just get the subset of data that you want.

- OPeNDAP servers allow this so you download or access what you need
- or we can work with data in cloud buckets using earthaccess Python package

Resources

- PACE HackWeek 2024 <https://pacehackweek.github.io/pace-2024/>
- ARSET Course
https://appliedsciences.nasa.gov/sites/default/files/2024-10/PACE_Part2_Final_0.pdf
- NOAA HackDays <https://nmfs-opensci.github.io/NMFSHackDays-2025/>

DEMO TIME!